

CAPSTONE PROJECT

Healthletics: AI-Powered Member Retention and Churn Prediction System

INTRODUCTION

Background

Healthletics is a fitness and wellness business focused on providing an inclusive environment where individuals with different backgrounds, fitness goals, and levels of experience can participate in structured training and wellness activities.

As Healthletics grows, effective member management and retention become increasingly important. In a traditional gym environment, member engagement and attendance may be observed through daily operations, but systematically identifying patterns that may indicate declining engagement or potential membership churn can be challenging without a structured, data-driven approach.

Artificial Intelligence (AI) and Machine Learning (ML) provide an opportunity to use member information and behavioral patterns to support better decision-making. Instead of relying solely on manual observation, a machine learning system can analyze historical member data and identify patterns associated with members who have previously churned.

Why This Domain Was Selected

The fitness and gym management domain was selected because it represents a real-world business environment where Artificial Intelligence and Machine Learning can provide practical value.

Healthletics was selected as the business context because the researcher is directly involved in the operation of the business. This provides an opportunity to connect the concepts learned throughout the AI and Machine Learning program with an actual business problem.

Rather than developing a purely theoretical machine learning application, this project aims to explore how data-driven methods can eventually support Healthletics in areas such as member engagement, retention, and management decision-making.

The use of Healthletics as the business context also provides a foundation for future development beyond this capstone project. The knowledge and findings gained from this study may eventually contribute to the development of a broader Healthletics membership and gym-management system.

Project Motivation

Member churn is an important concern in the fitness industry because losing existing members can affect membership growth, revenue, and long-term customer relationships.

The motivation for this project is therefore to investigate whether machine learning can identify patterns associated with gym-member churn and help Healthletics identify members who may be at higher risk of leaving.

For this capstone, a publicly available gym-member churn dataset is used to develop and evaluate the machine learning approach. The public dataset provides a practical starting point for demonstrating the end-to-end machine learning lifecycle while avoiding the use of private or sensitive Healthletics member information.

The resulting model is intended as a proof of concept, rather than an immediately deployable production system for Healthletics. Future development could involve collecting appropriate and anonymized Healthletics membership, attendance, and engagement data and using these data to further develop and validate the predictive system.

Project Direction

The long-term vision is to gradually develop Healthletics into a more data-driven and technology-enabled fitness business.

The capstone focuses specifically on member retention and churn prediction as an initial AI/ML application:

Member Data



Data Analysis



Machine Learning Model



Churn Risk Prediction



Retention Decision Support

In the future, this concept could be expanded by incorporating additional Healthletics data such as membership records, attendance, payments, class participation, membership renewals, and other relevant engagement information.

This approach allows the current capstone to remain focused, realistic, and achievable, while establishing a foundation for future Healthletics technology and AI initiatives.

STEP 1. PROBLEM UNDERSTANDING & FRAMING

1.1 Project Title

Healthletics: AI-Powered Member Retention and Churn Prediction System

1.2 Business Context

Healthletics Fitness Club is a fitness and training facility that provides services including gym training, CrossFit, HYROX, and group fitness activities. As the business grows, effective member management and retention become increasingly important.

One of the key challenges in gym management is understanding why members stop attending or fail to renew their memberships. Member churn can directly affect revenue, membership growth, resource planning, and long-term customer relationships. In a manually managed environment, identifying members who are becoming disengaged can be difficult, especially as the number of members increases.

A predictive analytics system can help Healthletics move from a reactive approach to a more proactive retention strategy. Instead of waiting until a member has already cancelled or failed to renew, the business can identify behavioral patterns associated with increased churn risk and potentially intervene earlier.

The long-term vision is to integrate this capability into a future Healthletics membership and gym management system that can combine membership, attendance, class participation, payment, and engagement information.

1.3 Business Problem Statement

Healthletics needs a data-driven method to identify gym members who are at higher risk of churning or not renewing their membership.

Currently, member retention decisions may rely heavily on manual observation and staff experience. This approach can make it difficult to consistently identify declining engagement across a growing membership base.

The business problem addressed by this project is therefore:

How can Healthletics use membership and activity-related data to identify members who are at risk of churn and support proactive member-retention strategies?

The objective is not only to predict which members are likely to churn, but also to understand the behavioral factors associated with churn so that Healthletics can design more targeted and timely retention interventions. Move from a primarily reactive approach to a more proactive, data-informed retention strategy.

1.4 Data Science Problem

The data science problem is to develop a machine learning model that can predict whether a gym member is likely to churn based on available membership, demographic, and activity-related characteristics.

The project uses historical member-level data containing information such as contract duration, lifetime, class attendance frequency, current-month activity, age, additional charges, group participation, and other membership-related variables.

The target variable is Churn, where:

- 0 = Retained member
- 1 = Churned member

The machine learning system therefore learns patterns from historical data and uses those patterns to classify members into retained or churned categories.

The analysis focuses on identifying behavioral and membership-related patterns associated with churn and developing a classification model that can distinguish between members who are likely to remain and those who may be at higher risk of churn.

The project therefore translates the business problem of member retention into a measurable machine learning prediction problem

1.5 Machine Learning Task Type

This project is a supervised learning — binary classification problem.

The target variable is Churn, with the following interpretation:

- 0 — Retained: the member does not churn.
- 1 — Churned: the member churns or does not renew.

The model uses available membership, demographic, participation, attendance-frequency, lifetime, and engagement-related variables to predict the target outcome.

The project evaluates multiple supervised classification algorithms and compares their performance to determine which model provides the most effective prediction of member churn.

Models Implemented

The following classification models are implemented and evaluated:

- Logistic Regression — used as a baseline classification model.
- Decision Tree — used to capture nonlinear relationships and decision patterns.
- Random Forest — an ensemble model used to improve predictive performance and provide feature-importance insights.

The models are evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC, with ROC-AUC serving as the primary evaluation metric.

1.6 Project Objective

The primary objective of this project is to develop and evaluate a machine learning proof of concept that can identify members at higher risk of churn and provide insights that may support proactive member retention strategies at Healthletics.

Specifically, the project aims to:

1. Understand the membership and activity characteristics associated with churn.
2. Prepare and analyze the available gym membership dataset.
3. Engineer meaningful behavioral features that may improve churn prediction.
4. Develop and compare supervised machine learning classification models.
5. Evaluate model performance using appropriate classification metrics.
6. Identify important signals associated with predicted churn risk.
7. Examine model behavior across selected demographic groups as part of responsible AI and fairness assessment.
8. Provide a foundation for a future Healthletics member management and analytics system.

The project also aims to provide information that can support future Healthletics retention strategies, such as identifying declining member engagement and prioritizing members who may require follow-up.

1.7 Model Success Metrics

Because the business objective is to identify members who may be at risk of leaving, model performance should not be evaluated using accuracy alone.

The project therefore uses the following evaluation metrics:

Primary Metric: ROC-AUC (Receiver Operating Characteristic—Area Under the Curve).

ROC-AUC is selected as the primary metric because it measures the model's ability to distinguish between retained and churned members across different classification thresholds.

This is particularly useful for a churn prediction system because the business may adjust the intervention threshold depending on the cost of missing a potentially at-risk member.

The following supporting metrics are also used:

- Accuracy — overall proportion of correctly classified members.
- Precision — proportion of members predicted as churners who actually churned.
- Recall — proportion of actual churners correctly identified by the model.
- F1-score — balance between precision and recall.

Recall is particularly important from a business perspective because failing to identify a member who is genuinely at risk of churning may result in a missed opportunity for timely retention support.

Because the dataset contains more retained members than churned members, evaluating multiple metrics provides a more complete assessment of model performance than accuracy alone.

Model performance will therefore be evaluated using a combination of ROC-AUC, recall, precision, F1-score, and accuracy.

1.8 Business KPIs

The primary business KPI for the proposed system is Member Retention Rate.

Supporting business KPIs include:

- Churn Rate

- Renewal Rate
- Number of At-Risk Members Identified
- Retention of Previously Identified At-Risk Members
- Average Member Attendance
- Member Engagement
- Potential Revenue Retained

These KPIs connect the machine learning output to measurable business outcomes.

The effectiveness of the proposed churn-prediction approach would ultimately be evaluated through measurable changes in retention and engagement outcomes during a future Healthletics pilot. The model itself does not guarantee improved retention; rather, it is intended to provide information that can support timely and appropriate retention actions.

1.9 Expected Business Value

A successful churn prediction system could allow Healthletics to:

1. Identify at-risk members earlier before they leave.
2. Prioritize retention efforts toward members with higher predicted churn risk.
3. Understand engagement patterns associated with member churn.
4. Improve member retention strategies using data rather than relying solely on manual observation.
5. Support future automated member management through integration with attendance, membership, payment, and class participation data.
6. Improve revenue stability by reducing avoidable membership losses.

The system is intended to function as a decision-support tool, rather than automatically making decisions about members.

1.9 Expected Business Value

The expected business value of the project is to provide Healthletics with a data-driven method for identifying potential churn risk earlier and prioritizing members who may benefit from retention support.

If validated using actual Healthletics member data, the proposed approach could support:

- Earlier identification of declining member engagement.
- More targeted retention outreach.
- Better understanding of member behavior and engagement patterns.
- More informed membership and retention planning.

- Improved ability to prioritize staff attention and retention resources.
- A foundation for future data-driven member management.
- Potential improvement in member retention and renewal outcomes.

The project does not claim a specific financial return because actual Healthletics financial and retention data were not used in this proof of concept. Future business value should be established through a controlled pilot using actual member data and measurable retention outcomes.

1.10 Project Scope and Data Context

This project uses a public gym membership dataset, Gym Customers Features and Churn, as a proof-of-concept dataset.

The dataset contains 4,000 member records and 14 variables covering membership characteristics, demographics, participation, attendance frequency, member lifetime, additional charges, and churn.

The dataset contains:

- 2,939 retained members (73.48%)
- 1,061 churned members (26.53%)

The dataset contains no missing values and no duplicate records based on the initial data-quality assessment.

The project scope includes:

- Problem definition and business framing.
- Data understanding and data-quality assessment.
- Exploratory data analysis.
- Feature engineering.
- Feature selection.
- Data standardization.
- Principal Component Analysis (PCA).
- Supervised machine learning model development.
- Model comparison and evaluation.
- Feature importance and model interpretation.
- Bias and fairness assessment across selected groups.
- Ethical AI considerations.
- Business interpretation and recommendations.

The project is not a production deployment of an operational Healthletics membership system. The dataset is publicly available and is therefore used to demonstrate the machine learning workflow and establish a proof of concept.

A future production implementation would require actual Healthletics member data, appropriate data governance and privacy controls, validation on local member populations, monitoring of model performance, and integration with the organization's membership and operational systems.

For this capstone, the analysis uses a publicly available gym customer churn dataset from Kaggle. The dataset provides historical member characteristics and activity-related information that can be used to demonstrate the feasibility of churn prediction.

The dataset is used as a proof-of-concept dataset for the Healthletics business problem. It does not represent actual Healthletics member records.

In a future Healthletics implementation, the same analytical approach could be applied to appropriately anonymized and governed Healthletics data, including membership information, attendance records, class participation, contract information, and other relevant engagement indicators.

1.11 Step 1 Summary

This step established the business and data science foundation for the Healthletics: AI-Powered Member Retention and Churn Prediction System.

The business problem is the need to identify potential member churn risk early enough to support proactive retention efforts. This was translated into a supervised binary classification problem, with Churn as the target variable.

The project uses a public gym membership dataset as a proof of concept and evaluates model performance using ROC-AUC as the primary machine learning metric, supported by accuracy, precision, recall, and F1-score.

From a business perspective, Member Retention Rate is identified as the primary KPI, supported by churn rate, renewal rate, engagement, attendance, and other retention-related measures.

The project is designed as an analytical proof of concept rather than a production system. Its ultimate value will need to be validated using actual Healthletics member data and measurable business outcomes.

With the problem, objectives, success metrics, business KPIs, and project scope established, the project can proceed to Step 2 — Data Collection & Understanding.

STEP 2. DATA COLLECTION & UNDERSTANDING

2.1 Dataset Selection and Source

For this project, a public gym-customer churn dataset titled “Gym Customers Features and Churn” dataset by Adrian Vinueza, obtained from Kaggle. It was selected as the basis for developing a proof-of-concept member churn prediction system for Healthletics.

The dataset was selected because its variables are relevant to the business problem identified in Step 1. It contains information related to membership contracts, member activity, engagement, demographic characteristics and churn status. These variables provide a suitable basis for investigating patterns associated with member retention and churn.

The dataset contains 4,000 member records and 14 variables, including 13 predictor variables and one target variable, Churn.

Because the dataset is a public dataset rather than actual Healthletics member data, the results should be interpreted as a proof of concept. Before any production implementation, the model would need to be retrained and validated using appropriately collected, anonymized Healthletics data.

Dataset: Gym Customers Features and Churn
Author: Adrian Vinueza

Source: Kaggle

Dataset URL: <https://www.kaggle.com/datasets/adrianvinueza/gym-customers-features-and-churn>

The dataset is used as a proof-of-concept dataset for the Healthletics capstone. It is not actual Healthletics member data.

2.2 Dataset Overview

The dataset contains information related to member demographics, membership characteristics, participation, activity frequency, and churn status.

The dataset consists of:

- 4,000 rows
- 14 columns
- 13 predictor variables
- 1 target variable: Churn

The target variable, Churn, is coded as:

- 0 = Retained
- 1 = Churned

The dataset contains:

- 2,939 retained members (73.48%)
- 1,061 churned members (26.53%)

This indicates that the target variable is not perfectly balanced, with retained members representing the larger group. This imbalance was considered when selecting and evaluating the machine learning models.

2.3 Data Dictionary

Variable	Data Type	Description
gender	Integer	Binary-coded gender variable
Near_Location	Integer	Indicates whether the member is located near the gym
Partner	Integer	Indicates partner-related membership status
Promo_friends	Integer	Indicates participation through a friends promotion
Phone	Integer	Binary phone-related indicator
Contract_period	Integer	Membership contract duration in months

Group_visits	Integer	Indicates participation in group visits
Age	Integer	Member age
Avg_additional_charges_total	Float	Average additional charges associated with the member
Month_to_end_contract	Integer	Number of months remaining in the member's contract
Lifetime	Integer	Member lifetime/tenure measure
Avg_class_frequency_total	Float	Average class attendance frequency over the total membership period
Avg_class_frequency_current_month	Float	Average class attendance frequency during the current month
Churn	Integer	Target variable indicating whether the member churned

The dataset contains 10 integer variables and 4 floating-point variables.

The binary variables are represented numerically as 0/1. The dataset documentation used for this analysis does not provide sufficient information to assign additional semantic labels to every binary coding, so the analysis retains the original numerical representation rather than assuming a specific meaning for each code.

2.4 Data Types and Feature Categories

For analysis, the variables can be grouped into several categories.

Demographic Feature

- Age
- Gender

Membership and Contract Features

- Contract_period
- Month_to_end_contract
- Lifetime

Engagement and Activity Features

- Avg_class_frequency_total
- Avg_class_frequency_current_month
- Group_visits

Membership/Business Indicators

- Near_Location
- Partner
- Promo_friends
- Phone
- Avg_additional_charges_total

Target Variable

- Churn

This structure provides a combination of demographic, membership, engagement, and behavioral information that is appropriate for investigating member churn

2.5 Data Quality Assessment

Before modeling, the dataset was examined for missing values, duplicate records, data types, distributions, and potential outliers.

Missing Values

The dataset contains no missing values across the 4,000 records and 14 variables. Therefore, no imputation method was required for missing data.

Duplicate Records

The dataset contains no duplicate records. Therefore, no duplicate rows were removed during preprocessing.

Data Types

The dataset contains:

- 10 integer variables
- 4 floating-point variables

The numerical representation of the variables is suitable for machine learning after the appropriate preprocessing and scaling steps.

2.6 Variable Ranges and Distribution

The observed ranges of the main numerical variables were examined to understand the structure of the dataset.

Variable	Observed Range
gender	0–1
Near_Location	0–1
Partner	0–1
Promo_friends	0–1
Phone	0–1
Contract_period	1, 6, 12 months
Group_visits	0–1The
Age	18–41
Avg_additional_charges_total	0.148–552.591
Month_to_end_contract	1–12
Lifetime	0–31
Avg_class_frequency_total	0–6.024
Avg_class_frequency_current_month	0–6.147
Churn	0–1

The Contract_period variable has three observed membership duration: 1,6, and 12 months

2.7 Initial Churn Distribution

The target variable was examined before model development.

Of the 4,000 members:

- 2,939 members were retained
- 1,061 members churned

This corresponds to approximately:

- 73.48% retained
- 26.53% churned

Although the class distribution is not severely imbalanced, the churn class represents a smaller proportion of the dataset. Therefore, accuracy alone would not be sufficient to evaluate the usefulness of a churn prediction model.

This supports the use of additional metrics such as precision, recall, F1-score, and ROC-AUC, as established in Step 1.

2.8 Initial Observations from the Dataset

Initial examination of the data revealed several potentially important relationships with churn.

Members with shorter contract periods showed substantially higher churn rates.

Churn by Contract Period

Contract Period	Number of Members	Churn Rate
1 month	2,207	42.3%
6 months	833	12.5%
12 months	960	2.4%

This indicates a strong relationship between membership contract duration and churn in the dataset.

Group participation also showed a difference:

- Members without group visits: approximately 33.0% churn
- Members with group visits: approximately 17.3% churn

These observations provided useful direction for the EDA and feature-engineering work performed in Step 3.

2.9 Initial Correlation Analysis

Correlation analysis was performed to identify variables associated with the target variable.

The strongest observed correlations with churn included:

Variable	Correlation with Churn
Lifetime	-0.438
Avg_class_frequency_current_month	-0.412
Age	-0.405
Contract_period	-0.390
Month_to_end_contract	-0.381

Avg_class_frequency_total	-0.250
Avg_additional_charges_total	-0.199
Group_visits	-0.175

The negative correlations indicate that higher values of these variables are generally associated with lower churn in this dataset.

However, correlation alone does not establish causation. These relationships were therefore investigated further through EDA, feature engineering, and machine learning in later steps.

2.10 Potential Outliers

Potential numerical outliers were examined using an interquartile-range approach.

The following variables showed observations that could be flagged as potential outliers:

- Age — approximately 32 observations
- Avg_additional_charges_total — approximately 29 observations
- Lifetime — approximately 192 observations
- Avg_class_frequency_total — approximately 16 observations
- Avg_class_frequency_current_month — approximately 8 observations
-

These observations were not automatically removed.

Extreme values may represent legitimate members or valid behavioral patterns rather than errors. Removing them without sufficient evidence could result in the loss of useful information.

Binary variables were also not treated as conventional outliers simply because they contain only 0/1 values.

The decision was therefore to investigate the data contextually rather than automatically delete observations.

2.11 Important Feature Relationships

Two particularly strong relationships were observed among predictor variables:

- Contract_period and Month_to_end_contract: approximately 0.973 correlation
- Avg_class_frequency_total and Avg_class_frequency_current_month: approximately 0.953 correlation

These relationships indicate that some variables contain closely related information.

This was considered during feature engineering and dimensionality-reduction analysis in Step 3

2.12 Dataset Limitations

Although the dataset is appropriate for demonstrating the proposed churn-prediction methodology, several limitations should be recognized.

First, the dataset is a public dataset rather than actual Healthletics member data. Therefore, model results should be interpreted as a proof of concept rather than as predictions of actual Healthletics members.

Second, the available variables may not capture all factors that influence gym membership retention. For example, the dataset does not provide the complete operational context that a real Healthletics system could eventually capture.

Third, some variables have strong relationships with one another, which may introduce redundancy.

Finally, the dataset represents historical observations rather than actual Healthletics member data, the results should be interpreted as a proof concept, and therefore any production implementation, model need to be retrained and validated using appropriately collected, anonymized Healthletics data.

2.13 Summary of Step 2

The dataset provides 4,000 gym-member records with 14 variables, including membership characteristics, activity behavior, demographic information, and a binary churn outcome.

The data-quality assessment found no missing values and no duplicate records. The target variable contains 2,939 retained members and 1,061 churned members.

Initial analysis identified meaningful relationships between churn and variables such as lifetime, recent class frequency, age, contract duration, and months remaining on contract. Potential outliers and highly correlated predictors were also identified for further investigation.

Overall, the dataset provides an appropriate foundation for the next stage of the project: data preprocessing, exploratory data analysis, feature engineering, feature selection, and dimensionality reduction.

STEP 3. DATA PREPROCESSING, Applied EDA & Feature Engineering

3.1 Data Preprocessing and Cleaning

Before developing the machine-learning models, the dataset was examined and prepared to ensure that the data was suitable for analysis and modelling.

The preprocessing workflow included:

1. Checking the dataset structure and data types
2. Checking for missing values
3. Checking for duplicate records
4. Examining numerical distributions and potential outliers
5. Reviewing unusual or potentially influential observations
6. Creating a domain-relevant engineered feature
7. Selecting informative features
8. Scaling numerical predictors where required
9. Applying Principal Component Analysis (PCA) for dimensionality reduction
- 10.

The dataset contained **0 missing values and 0 duplicate records**. Therefore, no imputation technique was required and no rows needed to be removed because of duplication. This allowed the analysis to proceed without introducing assumptions through missing-value replacement.

Potential outliers were investigated using the **Interquartile Range (IQR) method**. Potential outliers were retained rather than automatically removed because unusual values may represent legitimate member behaviour, such as particularly high attendance, spending, or membership lifetime. The preprocessing steps were implemented in the Jupyter/Google Colab notebook to support reproducibility. An **80/20 stratified train-test split** was subsequently used for the final modelling process.

3.2 Outlier Assessment

Potential outliers were investigated using the interquartile range (IQR) method.

Potential observations were identified in variables including:

- Age
- Avg_additional_charges_total
- Lifetime
- Avg_class_frequency_total
- Avg_class_frequency_current_month

However, the presence of an extreme value does not automatically indicate an incorrect observation.

For example, a member with a longer lifetime or unusually high activity may represent a legitimate customer rather than a data-entry error.

Therefore, the project did not automatically delete all IQR-flagged observations. Instead, potential outliers were assessed in the context of the business problem.

This approach avoids unnecessarily removing legitimate member behavior that may be useful for predicting churn. The identified observations were therefore retrained for subsequent analysis and modeling.

3.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the distribution of variables and identify relationships between member characteristics and churn.

The analysis examined:

- Target distribution
- Numerical feature distributions
- Churn by contract period
- Churn by group participation
- Correlations between predictors and churn
- Relationships between member engagement and churn

The EDA provided important insights into the behavioral differences between retained and churned members.

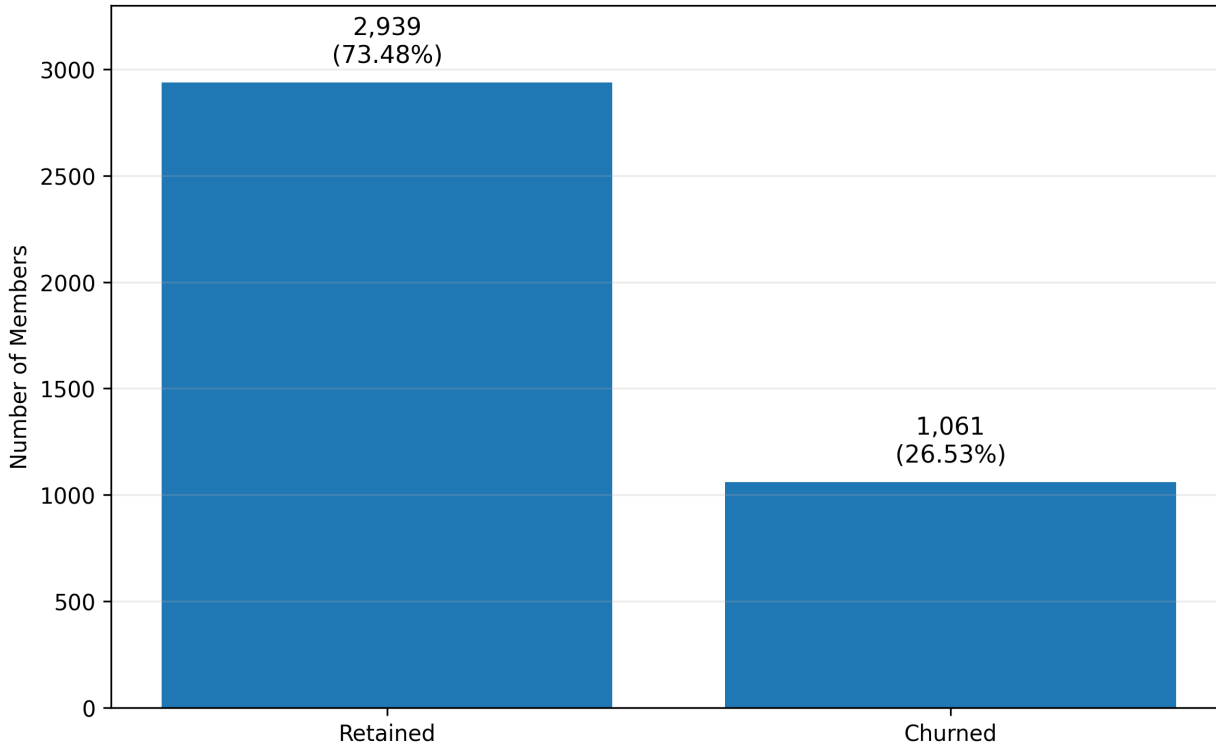
Key observations

Members who churned generally showed:

- shorter membership lifetime
- lower recent class frequency
- lower historical class frequency
- younger average age
- fewer months remaining on their contracts
- lower average additional charges

These observations provided a foundation for the feature-engineering stage.

Figure 3.1. Churn Distribution



3.4 Churn Distribution

The target variable was examined to determine the distribution of retained and churned members.

The dataset contained:

- 2,939 retained members — 73.48%
- 1,061 churned members — 26.53%

The target distribution was therefore moderately imbalanced.

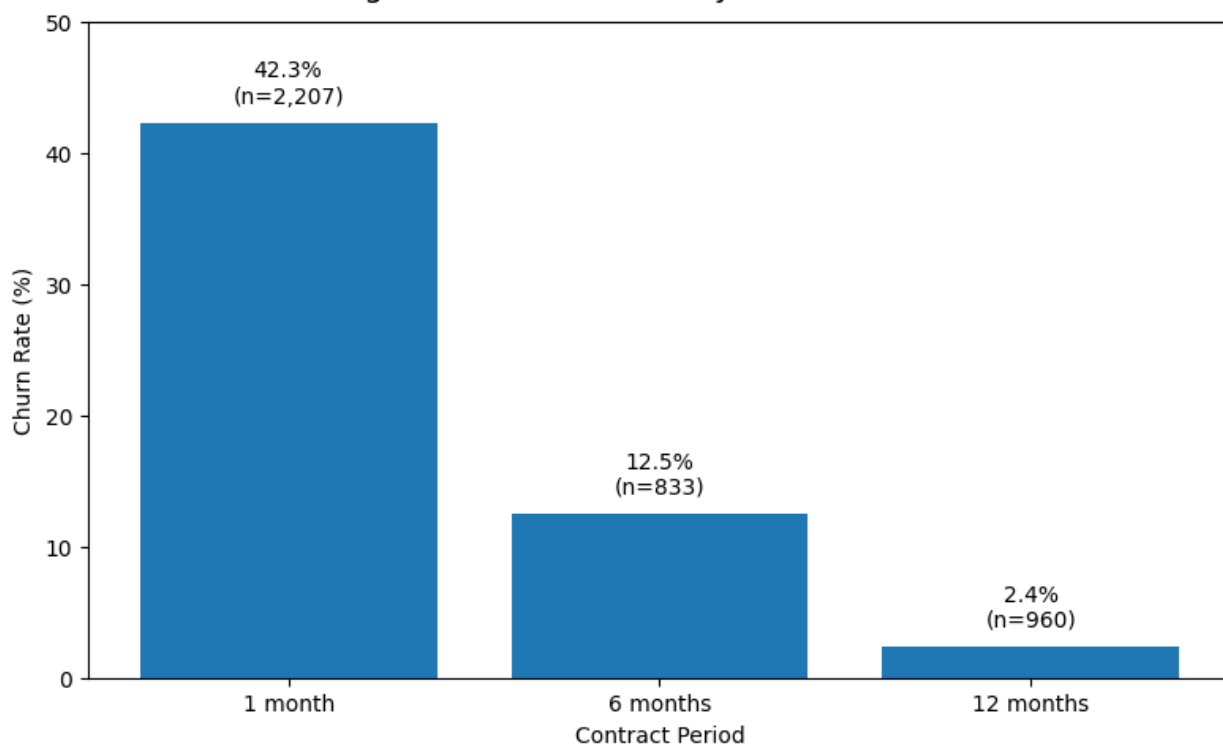
This finding influenced the model evaluation strategy. Accuracy was retained as a supporting metric, but the project placed greater emphasis on ROC-AUC, recall, precision, and F1-score.

3.5 Churn and Contract Duration

One of the strongest patterns identified during EDA was the relationship between contract duration and churn.

Contract Period	Number of Members	Churn Rate
1 month	2,207	42.3%
6 months	833	12.5%
12 months	960	2.4%

Figure 3.2. Churn Rate by Contract Period



The results show substantially higher churn among members with shorter contracts. Members with one-month contracts had a churn rate of approximately 42.3%, compared with approximately 2.4% among members with 12-month contracts.

This suggests that contract duration is an important variable for understanding member retention patterns within the dataset.

However, the observed relationship should not be interpreted as proof that longer contracts directly cause lower churn.

3.6 Churn and Group Participation

Group participation was examined to determine whether members who participated in group activities showed different churn patterns.

The observed churn rates were approximately:

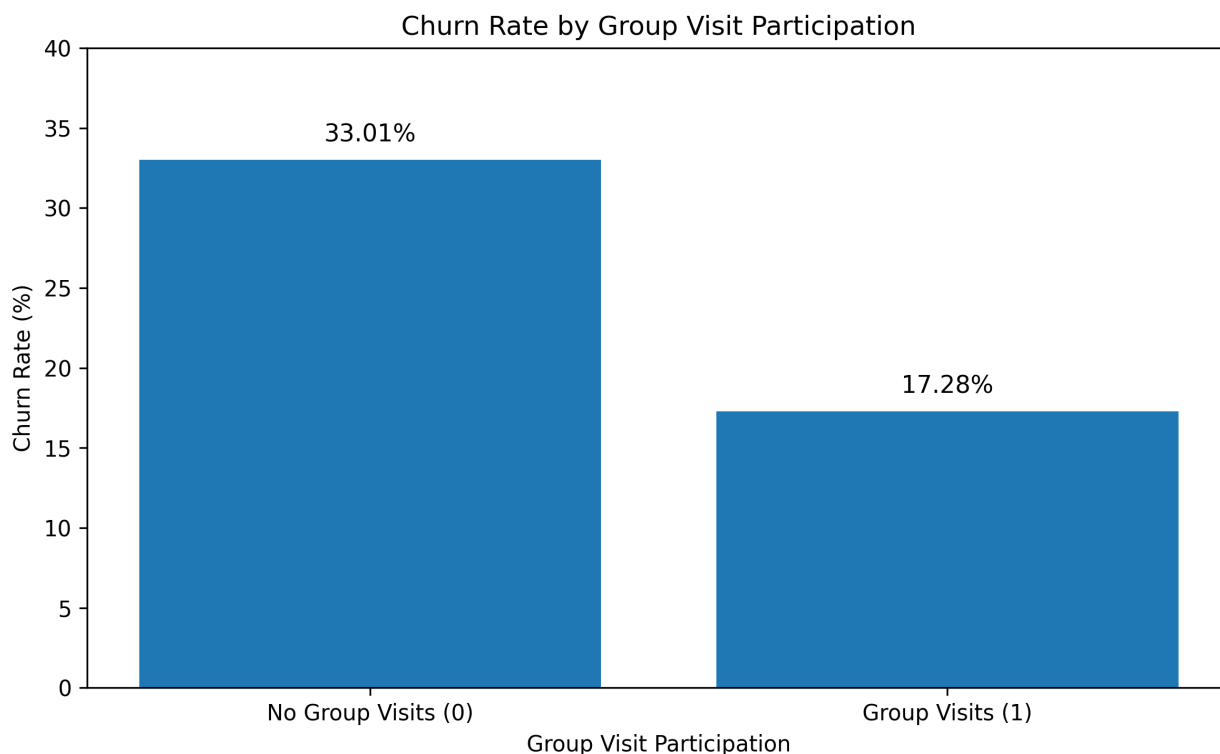
- 33.0% among members without group visits
- 17.3% among members with group visits

This indicates that members who participated in group activities had a substantially lower observed churn rate than members who did not participate in group visits.

This pattern may be associated with greater member engagement. Group-based activities may provide opportunities for social interaction, accountability, and continued engagement with the gym.

From a Healthletics business perspective, this finding may be relevant because group-based activities such as CrossFit, HYROX, and other community-based training are important components of the gym environment. Encouraging members to participate in appropriate group activities could therefore be considered as one potential element of a member-retention strategy.

However, the relationship observed in the dataset is associational rather than causal. Therefore, the analysis should not be interpreted as proof that group participation directly causes better retention.



3.7 Correlation Analysis

A correlation matrix was used to examine relationships among the numerical variables in the dataset and the target variable, Churn.

The strongest observed correlations with churn were:

Feature	Correlation with Churn
Lifetime	-0.438
Avg_class_frequency_current_month	-0.412
Age	-0.405
Contract_period	-0.390
Month_to_end_contract	-0.381
Avg_class_frequency_total	-0.250
Avg_additional_charges_total	-0.199
Group_visits	-0.175

The negative correlations indicate that higher values of these variables were generally associated with lower churn in this dataset.

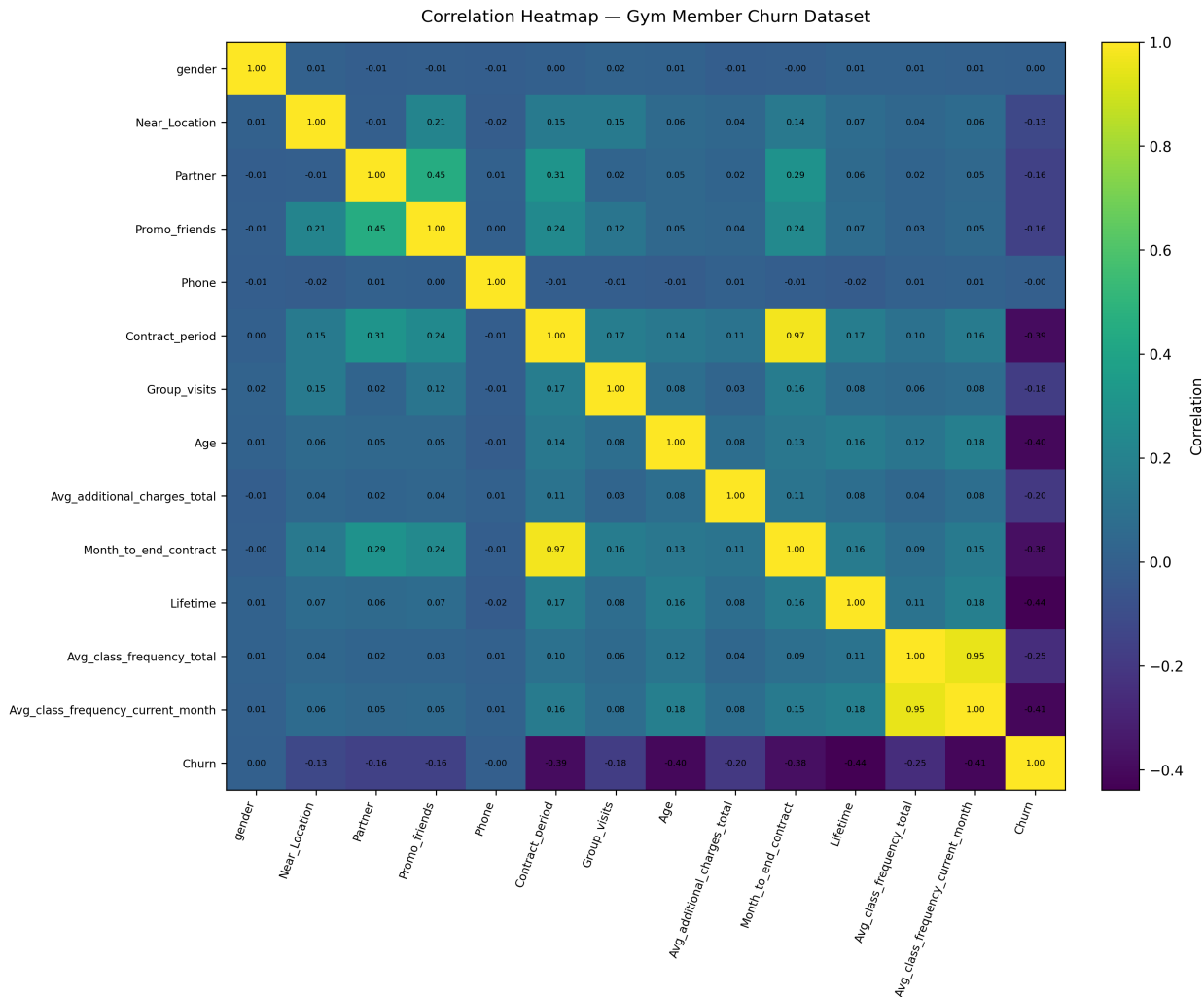
In particular, member lifetime and recent class attendance show relatively strong relationships with churn, supporting the earlier EDA findings.

Group_visits (-0.175) and Avg_additional_charges_total (-0.199) show weaker negative relationships with churn. Gender(0.001) and Phone (-0.001) show almost no linear relationship with churn.

The strongest relationships were associated with member tenure and activity frequency, supporting the importance of engagement-related variables in the churn prediction problem.

The correlation results provide an initial indication of potentially useful predictors. However, correlation alone does not establish causation, so the variables will be evaluated further using feature selection machine learning models.

Figure 3.4 — Correlation Matrix



3.9 Feature Engineering

A domain-relevant feature was created to capture changes in member activity over time. The purpose of this feature was to determine whether a member's recent activity was increasing or decreasing relative to their historical activity.

The engineered variable was:
frequency_change

It was calculated as:
frequency_change = Avg_class_frequency_current_month - Avg_class_frequency_tota

A negative value indicates that a member's current-month class frequency was lower than their historical average, while a positive value indicates relatively higher recent activity.

The feature was created to provide an additional measure of member engagement and changes in attendance behavior, which may be useful for predicting churn.

The engineered feature was retained for subsequent feature selection and machine-learning modeling.

However, the relationship observed in the dataset is associative and should not be interpreted as proof that a decline in attendance directly causes churn.

3.10 Frequency Change Results

The engineered feature showed a meaningful difference between retained and churned members.

The average frequency_change was approximately:

- Retained members: +0.003
- Churned members: -0.430
-

This indicates that churned members generally showed a decline in recent class frequency relative to their historical frequency, while retained members showed little overall change.

The correlation between frequency_change and churn was approximately:
-0.596

This was stronger than the correlations observed for the original individual attendance-frequency variables, indicating that the change in recent activity may provide useful information for identifying members at higher risk of churn.

The result supports the use of frequency_change as an informative feature for the subsequent machine-learning modeling stage.

Note: The relationship is observational and does not establish that a decline in attendance directly causes churn.

3.11 Handling Zero Activity Values

Zero values were identified in the class-frequency variables, including Avg_class_frequency_total and Avg_class_frequency_current_month.

The analysis identified:

- 88 members with a historical average class frequency of zero.
- 181 members with a current-month class frequency of zero.

These zero values were not treated as missing values or automatically removed, because a value of zero can represent a valid member behavior—specifically, that the member recorded no class attendance during the relevant period.

Removing these observations could eliminate meaningful information about inactive members, particularly because low or declining activity may be associated with higher churn risk

The zero values were therefore retained during preprocessing and incorporated into the feature-engineering analysis. The frequency_change feature was calculated using the difference between current-month and historical class frequency:

$$\text{frequency_change} = \text{Avg_class_frequency_current_month} - \text{Avg_class_frequency_total}$$

This approach allowed zero activity to remain part of the member engagement signal while avoiding unnecessary imputation.

The treatment of zero activity values was important because the project focuses on identifying patterns of declining engagement that may help predict member churn.

3.12 Feature Selection Using Mutual Information

Mutual information was used as a feature-selection approach to identify variables that contained useful information about the churn target.

The strongest mutual-information scores were:

Feature	Mutual Information
frequency_change	0.342321
Lifetime	0.208713
Contract_period	0.096917
Avg_class_frequency_current_month	0.095940
Month_to_end_contract	0.095073

Age	0.094497
Avg_class_frequency_total	0.058294
Avg_additional_charges_total	0.025032
Promo_friends	0.023650
Near_Location	0.013921
Group_visits	0.012424
Partner	0.010570
Phone	0.008077
gender	0.000000

Interpretation:

frequency_change produced the highest mutual-information score, followed by Lifetime. This indicates that these variables contained the greatest amount of information about the churn target among the evaluated features.

The result supports the earlier EDA findings that changes in member activity and membership tenure are important characteristics associated with churn.

Mutual information was used as a complementary feature-selection technique because it can identify informative relationships that may not be fully captured by simple linear correlation.

The results were considered when determining which variables should be retained for the subsequent machine-learning modelling stage.

However, mutual information represents statistical dependence and does not establish causation. A higher mutual-information score also does not guarantee that a feature will be the most important variable in every machine-learning model.

3.13 Feature Scaling

Numerical predictors were standardized before algorithms that are sensitive to feature scale.

Standardization was performed using the training data so that variables with different numerical ranges could be compared on a common scale.

The transformed training and testing datasets contained:

- Training set: 3,200 observations $\times$ 14 predictors
- Testing set: 800 observations $\times$ 14 predictors

The target variable was kept separate from the predictor variables.

Scaling was particularly relevant for the Logistic Regression model and for dimensionality-reduction analysis.

3.14 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was applied as a dimensionality-reduction technique to examine whether the information contained in the standardized predictor variables could be represented using fewer dimensions while retaining most of the variance in the dataset.

The analysis showed:

- 10 components explained approximately 91.13% of the variance.
- 11 components explained approximately 96.32% of the variance.
- 12 components explained approximately 99.81% of the variance.

Based on the objective of retaining more than 95% of the variance, 11 principal components were sufficient, retaining approximately 96.32% of the total variance.

PCA provided evidence that the information contained in the standardized predictor variables could be represented in a lower-dimensional space while retaining most of the information in the dataset.

This dimensionality-reduction analysis was used to assess the structure and redundancy of the predictor variables and to support decisions regarding the modelling stage.

Note: PCA was used as an analytical dimensionality-reduction assessment. The final model comparison was performed using the engineered predictor dataset rather than assuming that PCA components would automatically produce better predictive performance.

3.15 Preprocessing, EDA and Feature Engineering Summary

The preprocessing, exploratory data analysis, feature engineering, feature selection, and dimensionality-reduction stages produced several important findings:

1. The dataset contained no missing values and no duplicate records.
2. Potential outliers were investigated based on the numerical distributions. Legitimate observations were not automatically removed without contextual justification.
3. The churn target was moderately imbalanced, with retained members representing approximately 73.48% of the dataset and churned members approximately 26.53%. This supported the use of multiple evaluation metrics rather than relying on accuracy alone.
4. Exploratory analysis identified meaningful relationships between contract duration, member lifetime, attendance frequency, group participation, and churn.
5. Members with shorter lifetime and lower recent activity generally showed higher churn rates in the dataset.
6. The domain-relevant engineered feature `frequency_change` was created to measure the difference between a member's current-month class frequency and historical average frequency.
7. `frequency_change` showed a relatively strong negative correlation with churn (approximately -0.596) and produced the highest mutual-information score (0.342321) among the evaluated features.
8. Mutual information was used as a feature-selection technique to identify variables containing useful information about the churn target.
9. Numerical predictors were standardized to place variables with different scales on a comparable basis for subsequent modelling and PCA.
10. PCA was applied as a dimensionality-reduction assessment. The analysis showed that 11 principal components retained approximately 96.32% of the total variance.

Overall, the preprocessing and EDA stages provided a reproducible foundation for the subsequent machine-learning modelling stage. The analysis identified member engagement, activity change, and membership tenure as particularly relevant characteristics for understanding and predicting churn.

3.16 Step 3 Summary

The data-preparation process combined data-quality assessment, exploratory analysis, feature engineering, feature selection, and dimensionality reduction.

The most important feature-engineering result was the creation of `frequency_change`, which captured the difference between current and historical class frequency. The feature showed a substantially stronger relationship with churn than many of the original variables.

Mutual information was used for feature selection, while PCA was used for dimensionality reduction. The PCA analysis demonstrated that 11 components retained approximately 96.32% of the variance.

Overall, the preprocessing and EDA stage established a reproducible analytical foundation for model implementation and comparison in Step 4.

4.1 Model Implementation and Configuration

Following the preprocessing, exploratory data analysis, feature engineering, feature selection, and dimensionality-reduction stages, supervised machine-learning models were implemented to predict gym member churn.

The problem was formulated as a binary classification task, with Churn as the target variable, where 0 represents retained members and 1 represents churned members.

Three supervised classification algorithms were selected to provide complementary modeling approaches:

1. Logistic Regression — used as a baseline linear classification model.
2. Decision Tree — used to capture non-linear relationships and decision rules.
3. Random Forest — used as an ensemble tree-based model capable of capturing more complex relationships while reducing the limitations of relying on a single decision tree.

The models were trained using the prepared training dataset and evaluated on a held-out test dataset.

The model configurations used in the notebook included a Decision Tree with a maximum depth of 5 and a Random Forest with 200 estimators. These configurations were selected to provide appropriate model complexity while maintaining reproducibility. Model performance was evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC. Multiple metrics were used because the dataset contained more retained members than churned members, making it important to evaluate the model's ability to correctly identify churned members rather than relying on accuracy alone.

The modeling workflow was implemented in the Jupyter/Google Colab notebook to support reproducibility and comparison across models.

4.2 Train-Test Split and Modeling Preparation

The dataset was divided into training and testing subsets before model development to evaluate how well the trained models could generalize to unseen data.

An 80/20 train-test split was applied:

- Training set: 3,200 observations
- Testing set: 800 observations

The target distribution was maintained across the two subsets, with approximately 73.5% retained members and 26.5% churned members in both the training and testing data.

The predictor variables were then prepared for modeling. Numerical features were standardized for models that are sensitive to feature scale, particularly Logistic Regression. The same preprocessing approach was applied consistently to the training and testing data to avoid introducing information from the test set into model training.

The prepared training data were then used to develop the three classification models, while the held-out testing data were reserved for final model evaluation.

This approach provided a consistent and reproducible basis for comparing model performance across the different algorithms.

4.3 Logistic Regression

Logistic Regression was implemented as the baseline classification model for predicting gym member churn. It was selected because it provides a relatively simple and interpretable approach for binary classification and establishes a reference point for comparison with the tree-based models.

The model was trained using the prepared training data and evaluated using the held-out testing data.

The Logistic Regression model achieved the following results:

Metric	Result
Accuracy	0.9262
Precision	0.8844
Recall	0.8302
F1-score	0.8564
ROC-AUC	0.9775

The ROC-AUC score of 0.9775 indicates that the model was highly effective at distinguishing between retained and churned members in the test dataset.

The recall of 0.8302 indicates that the model correctly identified approximately 83% of the actual churned members.

Although Logistic Regression produced strong results, the model was subsequently compared with Decision Tree and Random Forest models to determine whether non-linear and ensemble approaches could provide improved predictive performance.

4.4 Decision Tree

A Decision Tree classifier was implemented to provide a non-linear alternative to the Logistic Regression baseline. Decision Trees can model interactions and non-linear relationships between predictor variables and the churn outcome through a series of decision rules.

The Decision Tree was configured with a maximum depth of 5. This configuration was used to control model complexity while allowing the model to capture meaningful patterns in the training data.

The model achieved the following results on the held-out test dataset:

Metric	Result
Accuracy	0.9463
Precision	0.9519
Recall	0.8396
F1-score	0.8922
ROC-AUC	0.9701

The Decision Tree improved upon Logistic Regression in terms of accuracy, precision, and F1-score. However, its ROC-AUC of 0.9701 was slightly lower than the Logistic Regression result of 0.9775, while recall increased only slightly from 0.8302 to 0.8396.

These results demonstrate that the Decision Tree was effective at classifying churn, but its overall performance did not exceed the other candidate models across the evaluation metrics. The model was therefore retained as an important comparison point before selecting the final model based on the complete performance comparison.

4.5 Random Forest

A Random Forest classifier was implemented as an ensemble-based approach to improve predictive performance by combining multiple decision trees. This approach was selected to provide a more robust non-linear model and to determine whether an ensemble method could improve upon the Logistic Regression and Decision Tree models.

The Random Forest model was configured with 200 estimators (decision trees). The model was trained using the prepared training dataset and evaluated using the held-out testing dataset.

The Random Forest achieved the following results:

Metric	Result
Accuracy	0.9650
Precision	0.9600
Recall	0.9057
F1-score	0.9320
ROC-AUC	0.9879

The Random Forest produced the strongest overall performance among the three evaluated models. In particular, its ROC-AUC of 0.9879 indicates strong discrimination between retained and churned members. Its recall of 0.9057 is also important for the business problem because the model correctly identified approximately 91% of the members who actually churned in the test dataset.

The model also achieved the highest accuracy, precision, recall, and F1-score among the three models evaluated.

Based on these results, Random Forest was identified as the best-performing candidate model for the churn prediction problem. The final model selection was based on the combined evaluation results rather than accuracy alone.

The model's feature importance results were subsequently examined to identify which member characteristics contributed most strongly to the model's predictions.

4.6 Model Performance Comparison

The three classification models were evaluated using the same held-out test dataset and the same performance metrics to enable a consistent comparison.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.9262	0.8844	0.8302	0.8564	0.9775
Decision	0.9463	0.9519	0.8396	0.8922	0.9701
Random Forest	0.9650	0.9600	0.9057	0.9320	0.9879

Comparison and Model Selection

The results show that Random Forest achieved the strongest overall performance across the evaluated models. It obtained the highest Accuracy (0.9650), Precision (0.9600), Recall (0.9057), F1-score (0.9320), and ROC-AUC (0.9879).

Random Forest's higher recall is particularly relevant to the business objective because identifying members who are actually at risk of churn is important for timely retention interventions.

Logistic Regression also performed strongly, with a ROC-AUC of 0.9775, demonstrating that the churn problem could be effectively modelled using a linear classification approach. The Decision Tree achieved higher accuracy and precision than Logistic Regression but had the lowest ROC-AUC (0.9701) among the three models.

Based on the combined evaluation results, Random Forest was selected as the best-performing model for the churn prediction task.

The model selection was therefore based on a combination of predictive performance measures, with particular attention to ROC-AUC, recall, and F1-score, rather than relying on accuracy alone.

Model Selection Rationale

The Random Forest model provides the strongest candidate for deployment in the proposed Healthletics churn prediction system because it combines:

- Strong overall predictive performance
- High ability to identify actual churners

- High discrimination between retained and churned members
- Strong precision and F1-score
- A suitable non-linear modelling approach for member behavior data

The selected model was then subjected to additional validation and interpretation to assess its stability and identify the factors most strongly associated with its predictions.

4.7 Cross-Validation and Reproducibility

To further assess the reliability of the selected Random Forest model, 5-fold cross-validation was performed using ROC-AUC as the evaluation metric. The resulting ROC-AUC scores were 0.9825, 0.9851, 0.9911, 0.9909, and 0.9902, with a mean ROC-AUC of 0.988 and a standard deviation of 0.0035.

The high mean ROC-AUC and low variation across folds indicate that the Random Forest model maintained strong and consistent classification performance across different subsets of the training data. This provides additional support for selecting Random Forest as the final candidate model.

For reproducibility, the notebook records the train-test split, preprocessing steps, feature set, and model configurations used during training. The Random Forest model was configured with 200 estimators, allowing the modelling process to be repeated using the same documented settings.

Key finding:

The Random Forest model achieved a mean cross-validated ROC-AUC of 0.988 ± 0.0035 , demonstrating strong and stable predictive performance.

Rubric connection:

Cross-validation strengthens confidence in model reliability, while documenting the preprocessing steps, feature set, and model configuration supports reproducibility.

4.8 Feature Importance and Model Interpretation

Feature importance was examined using the selected Random Forest model to identify which variables contributed most strongly to the model's churn predictions.

The most important features were:

Feature	Importance
frequency_change	0.3976
Lifetime	0.1946
Avg_class_frequency_current_month	0.0873
Age	0.0795
Avg_class_frequency_total	0.0624
Month_to_end_contract	0.0562

The results show that frequency_change was the most important feature, followed by member Lifetime. This supports the earlier EDA and feature-engineering findings that changes in recent activity and member tenure are strongly associated with churn.

The feature importance results provide an interpretable view of the model and help connect the machine learning results to practical retention strategies. However, feature importance represents the model's predictive contribution and should not be interpreted as proof of causation.

Key finding:

frequency_change accounted for the highest feature importance at 0.3976, indicating that changes in member activity were particularly influential in the Random Forest model's churn predictions.

4.9 Model Selection and Final Decision

Based on the evaluation results, the Random Forest model was selected as the final candidate model for the Healthletics churn prediction system.

Random Forest achieved the strongest overall performance across the evaluated metrics:

- Accuracy: 0.9650
- Precision: 0.9600
- Recall: 0.9057
- F1-score: 0.9320
- ROC-AUC: 0.9879

The model also achieved a mean cross-validated ROC-AUC of 0.988 ± 0.0035 , demonstrating consistent performance across different training subsets.

Random Forest was selected because it provided the best balance between correctly identifying churned members, minimizing incorrect predictions, and distinguishing between retained and churn-risk members. Its feature importance results also provide useful insight into the factors most associated with the model's predictions.

For the Healthletics business context, recall is particularly important because failing to identify a member who is likely to churn may represent a missed opportunity for timely retention intervention.

Final Step 4 decision:

The Random Forest model is the preferred model for the proposed Healthletics member retention and churn prediction system, subject to further validation and deployment considerations.

4.10 Step 4 Summary

Three supervised machine learning models were implemented for the Healthletics member churn classification problem: Logistic Regression, Decision Tree, and Random Forest. All models were evaluated using the same held-out test set and compared using Accuracy, Precision, Recall, F1-score, and ROC-AUC.

The Random Forest model achieved the strongest overall performance, with an Accuracy of 0.9650, Precision of 0.9600, Recall of 0.9057, F1-score of 0.9320, and ROC-AUC of 0.9879. It also achieved a mean cross-validated ROC-AUC of 0.988 ± 0.0035 , indicating stable performance across the validation folds.

Feature importance analysis identified frequency_change as the most influential feature, followed by Lifetime and recent class frequency. These findings were consistent with the earlier exploratory analysis and feature-engineering results.

Based on the comparative evaluation, cross-validation, and model interpretation, Random Forest was selected as the final candidate model for the proposed Healthletics member retention and churn prediction system.

Step 4 Outcome:

- Supervised classification models implemented and compared.
- Multiple evaluation metrics applied.
- Random Forest identified as the best-performing model.

- Cross-validation confirmed strong and consistent performance.
- Feature importance provided model interpretation.
- Model configurations and preprocessing steps were documented to support reproducibility.

STEP 5: CRITICAL THINKING → ETHICAL AI & BIAS AUDITING

5.1 Explainability and Model Interpretation

The final Random Forest model was examined further to improve the interpretability of its churn predictions. Feature importance and Partial Dependence Analysis (PDP) were used to understand which variables contributed most strongly to the model's predictions and how an important feature was related to predicted churn risk.

The Random Forest feature importance analysis identified `frequency_change` as the most influential predictor, followed by `Lifetime` and `Avg_class_frequency_current_month`. This indicates that changes in a member's recent activity relative to their historical activity provided important information for predicting churn risk.

The `frequency_change` feature was created as:

$$\text{frequency_change} = \text{Avg_class_frequency_current_month} - \text{Avg_class_frequency_total}$$

A negative value indicates that a member's recent class frequency was lower than their historical average, while a positive value indicates an increase in recent activity.

A Partial Dependence Plot (PDP) was used to further examine the relationship between `frequency_change` and the Random Forest's predicted churn probability. The PDP showed that predicted churn probability generally decreased as `frequency_change` improved from more negative values toward zero. This is consistent with the earlier exploratory analysis, where churned members showed a substantially lower average frequency change than retained members.

These findings provide a practical interpretation for Healthletics: a decline in recent member activity relative to a member's normal activity level can serve as an important signal of potential churn risk.

However, these findings represent predictive associations rather than causal relationships. A negative change in activity does not by itself prove that reduced attendance causes a member to churn. Other factors may also contribute to a member's decision not to renew.

5.2 Model and Data Limitations

Several limitations were considered when interpreting the results of the churn prediction model.

Class Imbalance

The dataset contains more retained members than churned members, with approximately 73.5% retained and 26.5% churned. Because of this imbalance, accuracy alone would not provide a sufficient assessment of model performance.

For this reason, multiple evaluation metrics were used, including precision, recall, F1-score, and ROC-AUC. Particular attention was given to recall because identifying members who are likely to churn is important for a retention-focused application.

Potential Feature Redundancy

The exploratory analysis identified strong relationships between some predictors. For example, `Contract_period` and `Month_to_end_contract` were highly correlated, as were `Avg_class_frequency_total` and `Avg_class_frequency_current_month`.

These relationships indicate that some variables contain overlapping information. Feature engineering, mutual information analysis, standardization, and PCA were therefore used during preprocessing and analysis to better understand the structure and information contained in the dataset.

Overfitting Consideration

The Random Forest achieved strong performance on the held-out test set, with a ROC-AUC of 0.9879. Cross-validation also produced a mean ROC-AUC of approximately 0.988 with a standard deviation of 0.0035, indicating consistent performance across the validation folds.

Nevertheless, strong predictive performance on this dataset should not automatically be interpreted as guaranteed performance on future Healthletics members. The model should be validated on new, real-world Healthletics data before being used operationally.

Data Scope and Generalizability

The dataset is a public gym-member dataset and may not fully represent the actual membership behavior of Healthletics members. Differences in membership pricing, programs, attendance patterns, coaching environment, demographics, and member motivations may affect how well the model generalizes to the Healthletics setting.

Therefore, the model should initially be treated as a proof of concept and analytical framework, with future validation using actual Healthletics membership and attendance data.

5.3 Bias and Fairness Analysis

A group-level bias and fairness analysis was performed to examine whether churn patterns and Random Forest predictions differed across relevant member groups. Gender and age were considered because they are demographic characteristics available in the dataset.

Gender Analysis

The observed churn rates were very similar between the two gender-coded groups:

Gender Group	Retained (%)	Churned (%)
0	73.51	26.49
1	73.44	26.56

The difference in observed churn rate between the two groups was very small, approximately 0.07 percentage points. Based on this dataset, there was no substantial difference in overall churn rate between the two gender-coded groups.

Because the dataset does not provide the semantic meaning of the numerical gender codes, the groups are reported as Gender 0 and Gender 1 rather than assigning a specific gender label.

Age-Group Analysis

Members were divided into four age groups:

- 18–24
- 25–29
- 30–34
- 35–41

The observed churn rates differed substantially across these groups:

Age Group	Retained (%)	Churned (%)
18–24	33.77	66.23
25–29	63.98	36.02
30–34	88.23	11.77
35–41	97.88	2.12

The youngest group had the highest observed churn rate at 66.23%, while the 35–41 group had the lowest at 2.12%. These differences indicate that age is strongly associated with churn in this dataset. However, the differences should not automatically be interpreted as evidence of discrimination because they may reflect underlying differences in membership behavior, contract patterns, engagement, or other characteristics.

Random Forest Performance by Age Group

The Random Forest was also evaluated separately across the age groups using accuracy, precision, and recall.

Age Group	Members	Accuracy	Precision	Recall
25–29	377	0.9655	0.9606	0.9385
30–34	328	0.9543	0.9091	0.7143
35–41	36	1.0000	0.0000	0.0000
18–24	59	1.0000	1.0000	1.0000

The results show that model performance was not identical across all age groups. The 25–29 group showed strong performance, while the 30–34 group had a lower recall of 0.7143.

The 35–41 group requires particular caution. Although its accuracy was 1.0000, its precision and recall were both 0.0000. This apparently contradictory result is likely related to the small number of test members in this group (36 members) and the very low observed churn rate. Therefore, the accuracy value alone should not be interpreted as evidence of excellent model performance for this group.

Similarly, the 18–24 group contained only 59 test members, so its perfect scores should also be interpreted cautiously.

Predicted Churn Rate by Age Group

The model’s predicted churn rates also differed across age groups:

Age Group	Members	Predicted Churn Rate (%)
18–24	59	67.80
25–29	377	33.69
30–34	328	10.06
35–41	36	0.00

The predicted churn rate was highest for members aged 18–24 at 67.80%, followed by 25–29 at 33.69%, 30–34 at 10.06%, and 35–41 at 0.00%.

These results show that the model’s predictions are not distributed equally across age groups. However, this difference should be interpreted carefully because the underlying churn rates also differ substantially across the age groups.

Overall, the fairness analysis demonstrates that model performance and prediction rates can vary across groups. These differences do not by themselves establish that the model is biased or discriminatory. They highlight the need for additional validation using larger and more representative data before deployment.

5.4 Fairness Considerations and Mitigation

The fairness analysis indicates that demographic groups can have substantially different churn patterns and model outcomes. Therefore, fairness should be treated as an ongoing monitoring requirement if the system is eventually deployed at Healthletics.

1. Review the Features Used

Features should have a clear business justification and represent legitimate indicators of member engagement or membership behavior. Demographic variables should be evaluated carefully before being used in automated decisions.

2. Monitor Performance Across Groups

Healthletics should continue monitoring model performance separately across relevant groups using measures such as:

- Accuracy
- Precision
- Recall
- False-positive rate
- False-negative rate
- Prediction rates

This can help identify whether a group is systematically receiving less accurate churn predictions.

3. Review Prediction Thresholds

If future validation identifies unacceptable differences in false-positive or false-negative rates between groups, prediction thresholds can be reviewed. Threshold decisions should consider both the business objective and fairness considerations rather than simply maximizing overall accuracy.

4. Human Oversight

The model should be used as a decision-support tool rather than an automatic decision-maker. A high churn-risk prediction should create an opportunity for appropriate member engagement, such as checking in with the member or offering support, rather than triggering an automatic negative action.

5. Continuous Monitoring

As Healthletics collects more real membership and attendance data, model performance and fairness should be reassessed periodically. Changes in member demographics, programs, pricing, or gym operations may affect model behavior over time.

5.5 Ethical AI Considerations

The Healthletics churn prediction system is intended to help identify members who may benefit from timely engagement before they discontinue their membership.

The model should therefore be used responsibly and transparently. A churn prediction represents risk, not certainty. A member identified as high risk should not automatically be assumed to have decided to leave.

Predictions should support positive interventions such as member check-ins, coaching support, engagement activities, and personalized retention initiatives rather than restricting membership benefits or services.

The current analysis is a proof of concept based on a public gym-member dataset. Before deployment at Healthletics, the model should be retrained and validated using appropriately collected and protected Healthletics membership and attendance data.

Responsible handling of member information should also include appropriate controls for data privacy, access, storage, and use.

5.6 Step 5 Summary

The ethical AI and bias analysis added an additional layer of evaluation beyond overall model performance.

The selected Random Forest model was interpreted using feature importance and Partial Dependence Analysis, with frequency_change identified as the strongest predictive feature.

Model and data limitations were also considered, including class imbalance, correlated predictors, potential overfitting, and the limitations of using a public dataset as a proxy for future Healthletics members.

A group-level bias and fairness analysis was performed using gender and age groups. Gender groups showed very similar observed churn rates, while age groups showed substantial differences in both observed churn and model predictions. Random Forest performance also varied across age groups, particularly in recall.

The analysis showed why fairness evaluation should consider multiple metrics and group sizes rather than relying on accuracy alone. The very small test samples for the 18–24 and 35–41 groups also demonstrate the importance of using larger and more representative data before deployment.

Finally, fairness mitigation strategies were identified, including feature review, group-level performance monitoring, threshold assessment, human oversight, and continuous monitoring.

Overall, Step 5 demonstrates that the proposed Healthletics churn prediction system should be evaluated not only on predictive performance, but also on explainability, fairness, responsible use, and suitability for real-world deployment.